LESSON PLAN

MOCKUP SQUADRON ENVIRONMENT (MSE) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	21	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Measurement of temperature using Wheatstone bridge, ratio meter coil and thermocouple.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Measurement of temperature using Wheatstone bridge, ratio meter coil and thermocouple.
5. Specific Objectives	24.2.1.1 Trainee will be able to measure the change in temperature using different methods of temperature measurement.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Wheatstone bridge setup for temperature measurement; Safety: Ensure power off before connecting probes; Instructor supervision: Monitor trainee connections and verify zero balance.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a Wheatstone bridge?
(b) How does a thermocouple generate an electrical signal?
(c) Why is a ratio meter coil used in temperature measurement?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Principles of Wheatstone bridge operation • Temperature measurement using Wheatstone bridge • Role of ratio meter coil in detection • Calibration of bridge for temperature changes • Error sources in bridge measurements	L & Ds	PPT, IB	15 min
(b)	• Thermocouple construction and types • Seebeck effect and voltage generation • Connection methods for thermocouples • Calibration of thermocouple sensors • Interference and noise reduction techniques	L & Ds	PPT, PDF	20 min
(c)	• Comparative analysis of measurement methods • Selection criteria for specific applications • Maintenance of temperature sensing devices • Troubleshooting common faults • Recording and interpreting temperature data	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a Wheatstone bridge detects temperature changes.
(b) Describe the working principle of a thermocouple.
(c) List two advantages of using a ratio meter coil.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between Wheatstone bridge and thermocouple methods for temperature measurement, referencing LP 02 concepts.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Measurement Of rotational speed using tacho generator and tacho probe.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min